

Task 5: Capture and Analyze Network Traffic Using Wireshark

Objective

To capture live network traffic using Wireshark, identify different network protocols, analyze packet details, and export the captured traffic as a .pcap file.

Tool Used

- Wireshark

System Configuration

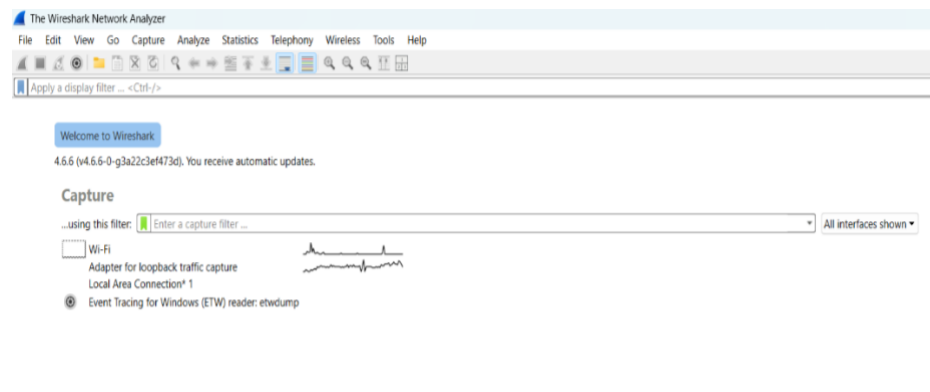
Parameter	Value
Operating System	Windows 11
Tool	Wireshark
Capture Type	Live Network Traffic Capture
File Format	.pcapng

Procedure

Step 1: Install Wireshark

Installed Wireshark and launched the application.

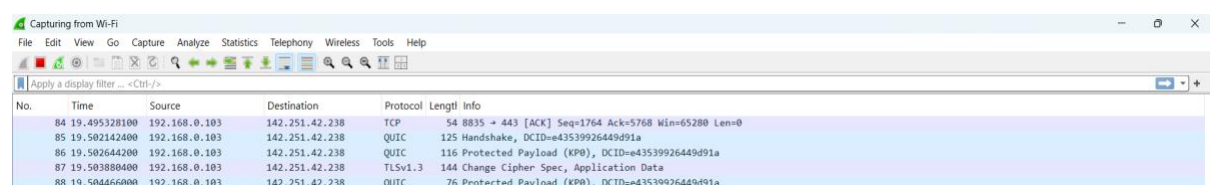
Screenshot 1: Wireshark Home Screen



Step 2: Select Active Network Interface

Selected the active Wi-Fi network interface for packet capture.

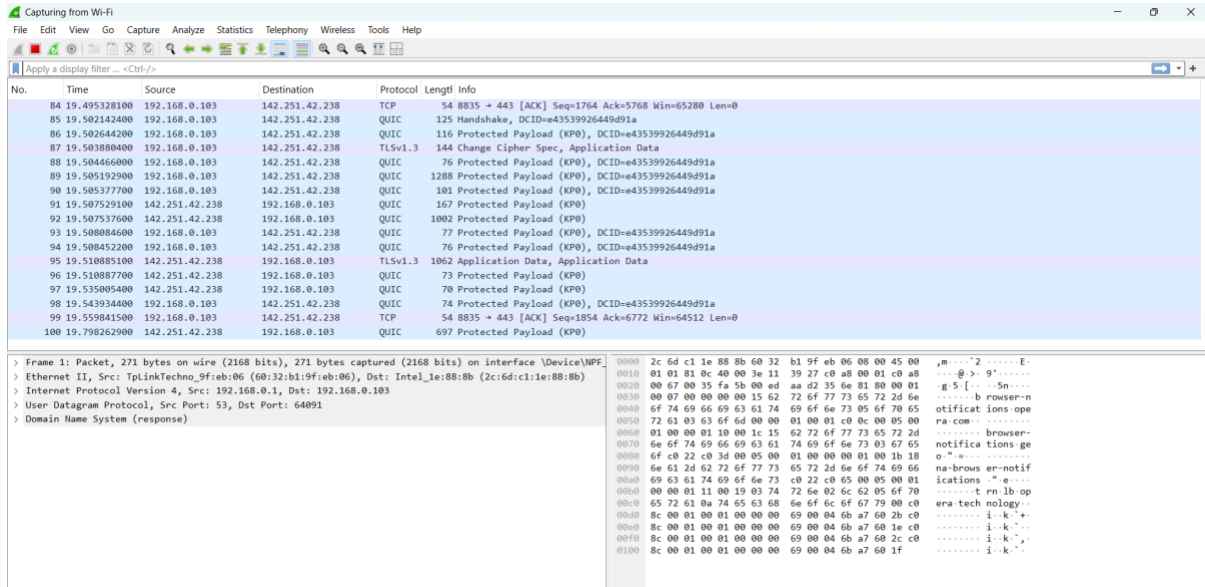
Screenshot 2: Active Network Interface



Step 3: Start Packet Capture

Started capturing live network packets.

Screenshot 3: Live Packet Capture Running



Step 4: Generate Network Traffic

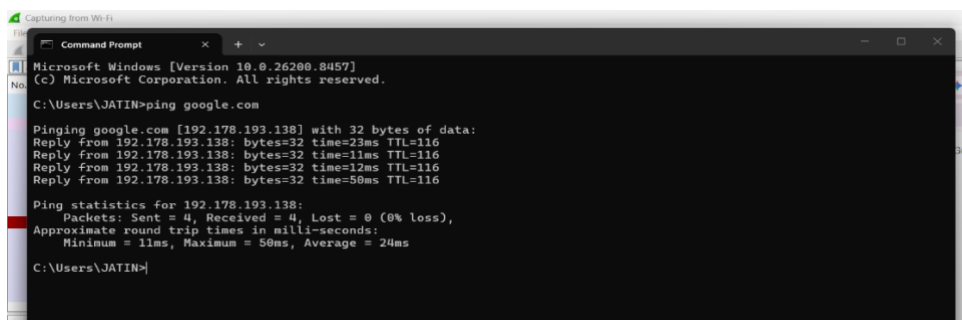
Generated network traffic by:

- Browsing websites such as Google, YouTube, and GitHub.
- Using ping command to communicate with external servers.

Command Used:

ping google.com

Screenshot 4: Generated Network Traffic



Step 5: Stop Packet Capture

Stopped packet capture after approximately one minute.

Screenshot 5: Captured Packets

Step 8: Filter TCP Packets

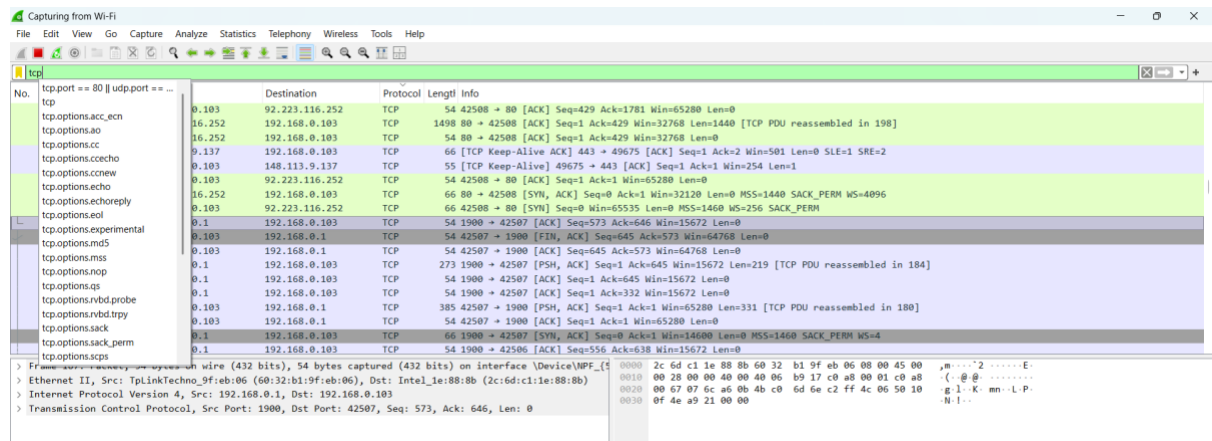
Filter Used:

tcp

Purpose:

To display Transmission Control Protocol packets.

Screenshot 8: TCP Filter Results



Step 9: Filter ICMP Packets

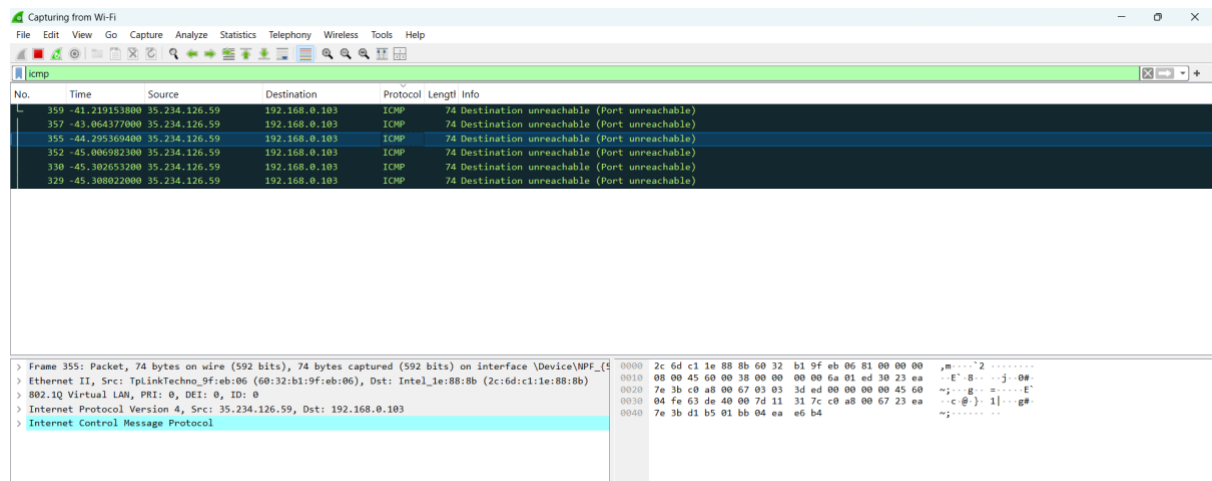
Filter Used:

icmp

Purpose:

To display encrypted HTTPS communication.

Screenshot 9: ICMP Filter Results

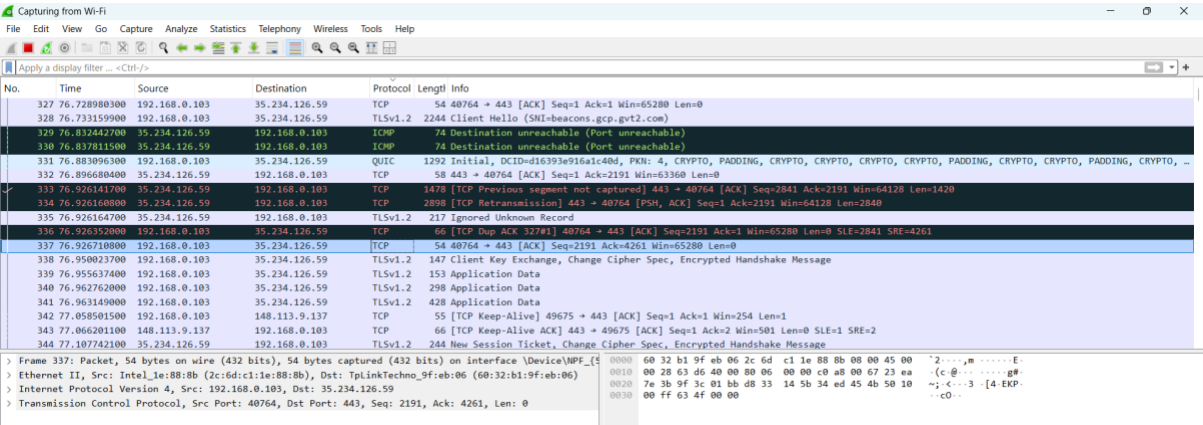


Step 10: Inspect Packet Details

Selected individual packets and examined:

- Source IP Address
- Destination IP Address
- Protocol Information
- Packet Length

Screenshot 10: Packet Details



Protocol Analysis

DNS (Domain Name System)

DNS is responsible for translating domain names into IP addresses.

Example:
google.com → IP Address

TCP (Transmission Control Protocol)

TCP provides reliable communication between devices and ensures packets are delivered correctly.

Findings

1. DNS traffic was observed during domain name resolution.
2. TCP packets established communication sessions.
3. TLS packets indicated encrypted HTTPS web traffic.
4. Packet details such as source and destination IP addresses were successfully analyzed.
5. Network traffic was successfully captured and exported.

Result

Successfully captured and analyzed live network traffic using Wireshark. Identified DNS, TCP, and TLS protocols and exported the capture as a .pcapng file for further analysis.